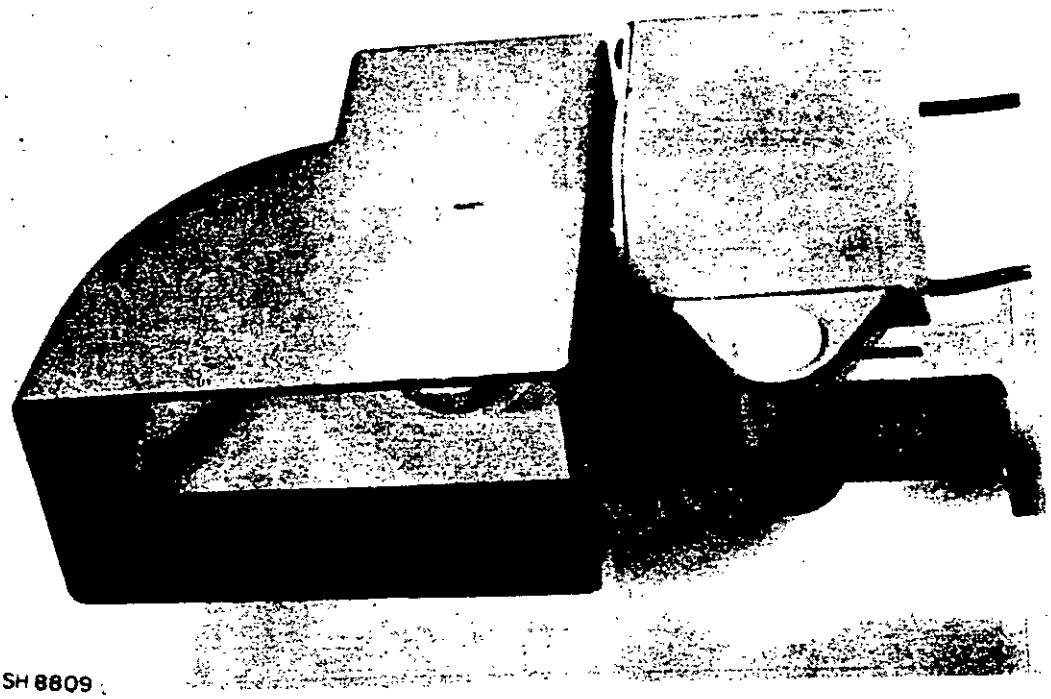
MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, DETAIL LIGHTING MINESWEEPER BUNKS, 15 WATTS, 120 VOLTS, NONMAGNETIC, SYMBOLS 5.1 AND 8.1

(A)

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).



SH 8809

FIGURE 1. Typical view of Symbol 5.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) denotes changes.

FSC 6210

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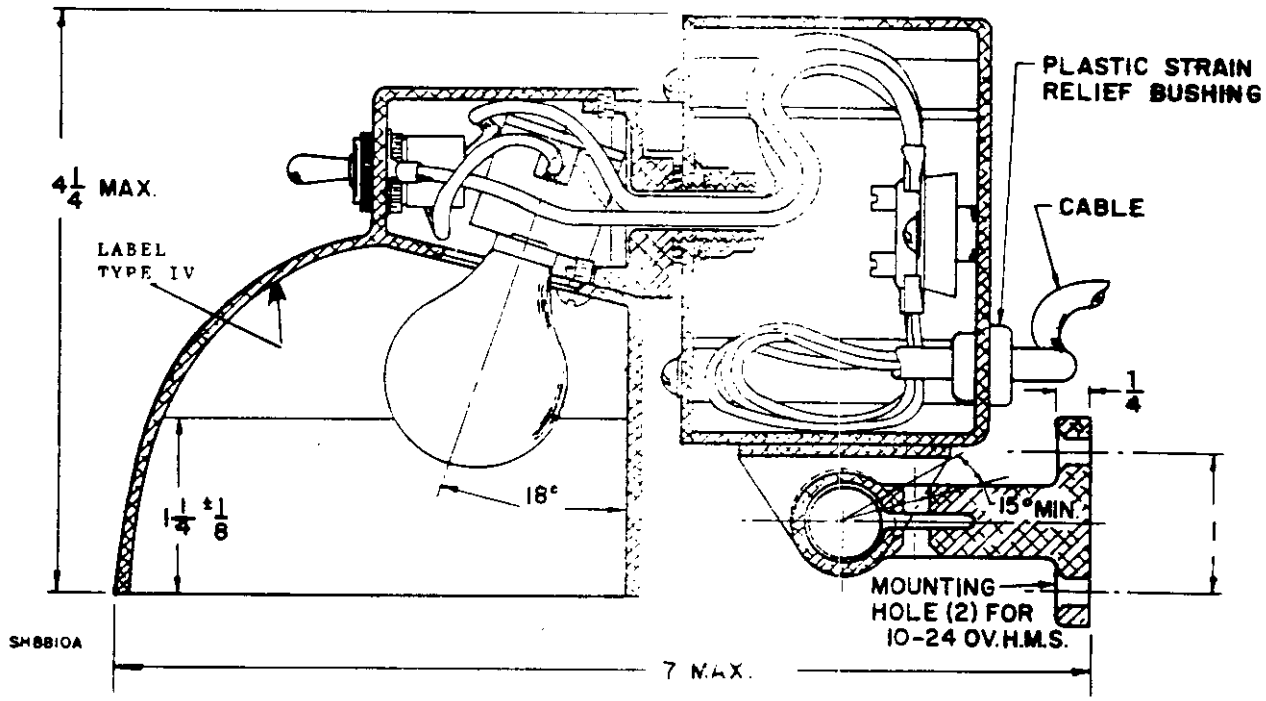


FIGURE 2. Typical section for Symbol 5.1.

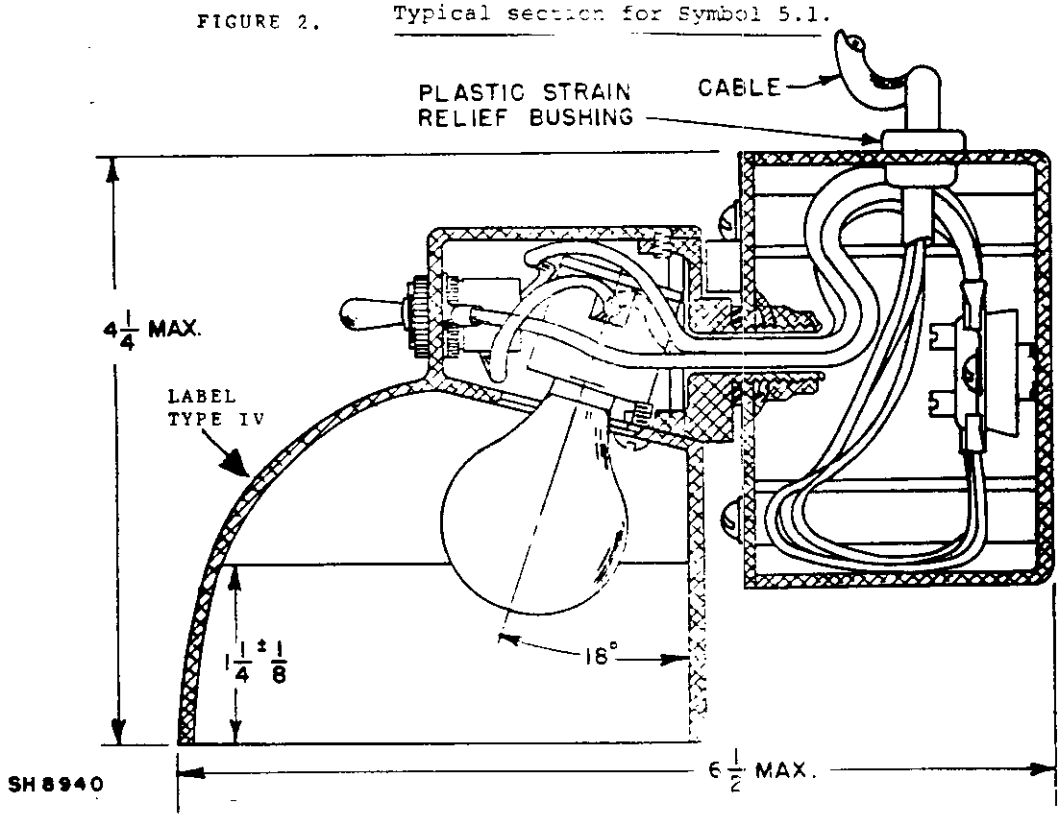


FIGURE 3. Typical section for Symbol 5.1.

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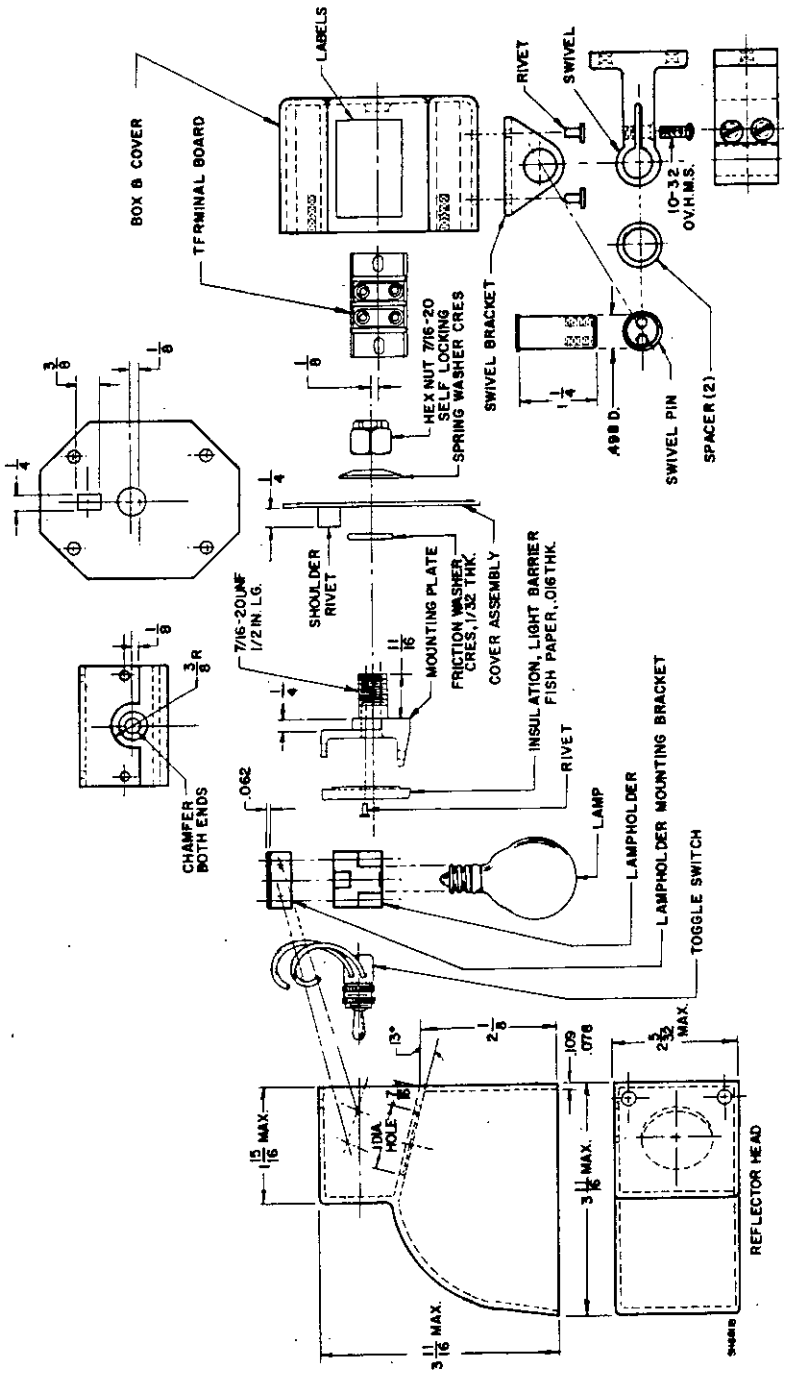


FIGURE 4. Parts assembly.

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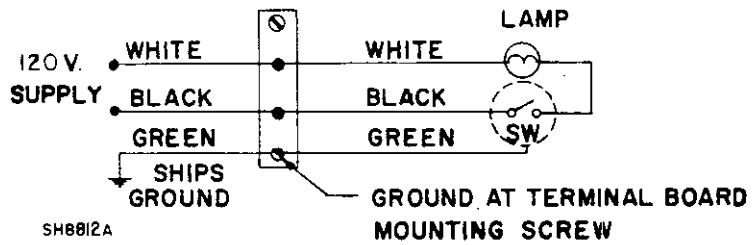
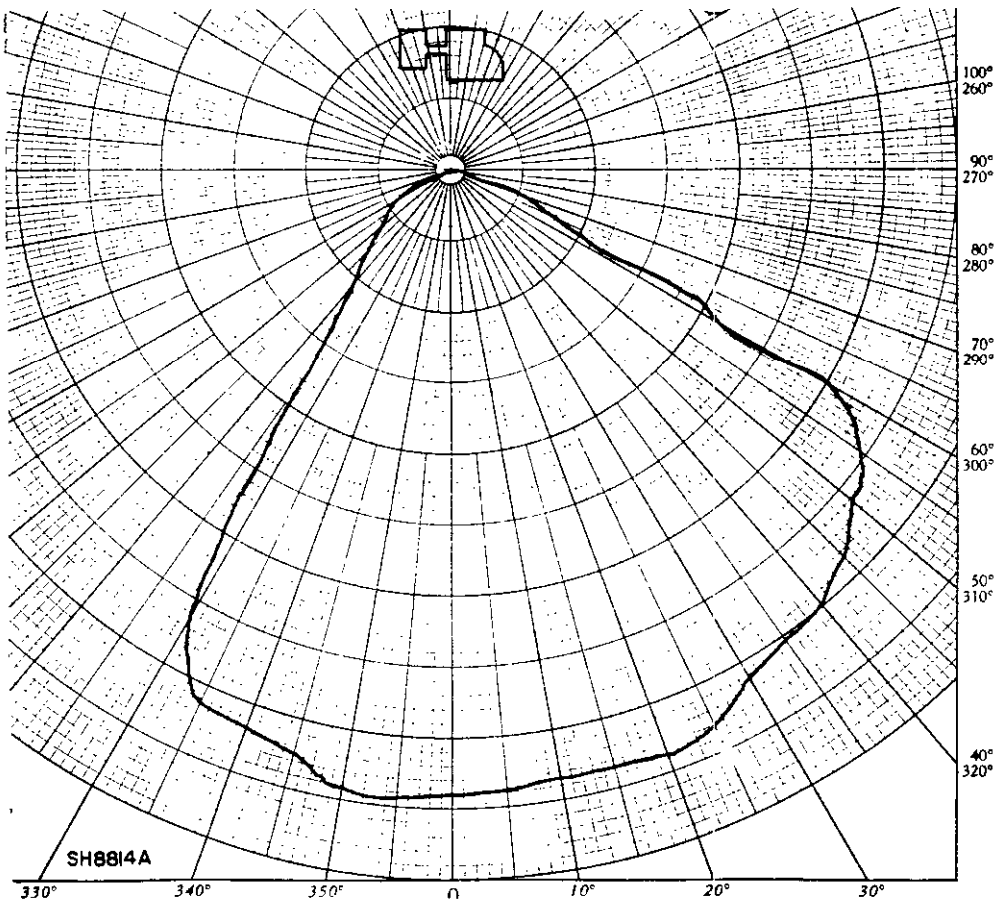


FIGURE 5. Wiring diagram.



Photometric data:
Distance: 15 inches.
Volts: 120.
Temperature: 78°F.
Plane: Rotated at filament of lamp.
Test shall be conducted with one 15-watt, (clear white) incandescent lamp covered by MS15575-2.

FIGURE 6. Candlepower distribution curves.

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REQUIREMENTS:
Type: II
Class: 1
Lights, bunk, for use with 15-watt clear (white) incandescent lamp, as follows:
Symbol 5.1: Bracket mounted.
Symbol 8.1: Box mounted.
Design:
Electrical rating: 120 volts, 15 watts.
Dimensions and configuration: See figures 1, 2, 3, and 4.
Material: All metal parts shall be nonmagnetic.
Parts assembly: See figure 4.
Box and cover: MIL-J-24142/8.
Terminal board: Four terminal, commercial grade.
Swivel pin: Nylon.
Lampholder: MIL-L-970/3.
Toggle switch: MS16656-1. The metal neck of the toggle switch shall be grounded. A green wire with wire terminals shall be used to connect the metal neck of the toggle switch and one of the mounting screws of the terminal board.

(A) Cable: 10 feet of type SVT construction in accordance with J-C-580 except AWG size 18, 3 conductor (white, black, and green.)Gray thermoplastic jacket, in accordance with J-C-580.

Bushing, strain relief: Heyco Manufacturing Company No. SR-5P-1, or equal.
Reflector heads:
Symbols 5.1 and 8.1: The inner surface extending upward 1-1/4 inches from the bottom edge shall be painted dull black. The reflector head shall rotate through a horizontal plane 90 degrees each side on a line perpendicular to connection box. The reflector heads shall have the following adjustment:
Symbol 5.1: Horizontal and vertical adjustment. The entire light shall swivel at least 15 degrees above a horizontal line through the swivel pin.
Symbol 8.1: Horizontal adjustment only.
Internal wiring: High temperature (105°C.) min.
Wiring diagram: See figure 5.
Distribution curve: Candlepower (see figure 6).
Lamp: Lights shall be designed to accommodate a 15-watt clear (white) incandescent lamp covered by MS15575-2, Industry Number 15S11/14 (Lamp is not furnished).
Weight:
Symbol 5.1: 1 pound, 15 ounces.
Symbol 8.1: 1 pound, 9 ounces.
Shoulder rivet: The shoulder rivet shall provide a positive stop of the reflector head and shall be of sufficient strength to withstand a minimum twist torque of 300 inch pounds.
Label: Label shall be furnished in accordance with MIL-F-16377. One type IV label shall be applied as shown on figures 2 and 3. The information to be shown on the type IV label shall be:

"INSTALL 15-WATT LAMP ONLY
NSN6240-00-797-8830"

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	See table I
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

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TABLE I. First article and comparison inspection.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Totally enclosed
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Torque		1/	
Grounding circuit	3.6.13.3	4.8.11	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

1/ The fixture shall be subjected to a torque test to determine conformance with the shoulder rivet requirement.

General Information: (a) Assignment of Military part numbers and general information is shown in table II.

TABLE II. Part number and general information.

Military Part number M16377/37-	Item name	National Stock Number (NSN)	Supersedes part number
5.1	Complete fixture, Symbol 5.1	6210-00-411-6024	9000-S6401-74070, Symbol 5.1
8.1	Complete fixture, Symbol 8.1	6210-00-411-6025	9000-S6401-74070, Symbol 8.1

Preparing activity:
Navy - SH
(Project 6210-N585)

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